

Task 1 Report: Machine Learning Classification

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Internship: InternSpark AI Internship

Task Number: Task 1

Project Title: Machine Learning Classification

1. Objective

The objective of this project is to build and evaluate a Machine Learning classification model capable of predicting target classes accurately using a supervised learning approach.

2. Problem Statement

Machine learning classification is widely used to automate prediction tasks by learning patterns from historical data. This project demonstrates the complete workflow of data preprocessing, model training, evaluation, and prediction.

3. Tools & Technologies

- **Languages:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- **Environment:** Jupyter Notebook / VS Code

4. Dataset

The dataset was preprocessed by handling missing values, encoding categorical variables where required, and splitting the data into training and testing sets.

5. Methodology

- Import required libraries.
- Load the dataset.
- Perform data preprocessing.
- Split the dataset into training and testing sets.
- Train the Machine Learning classification model.
- Evaluate the model using standard classification metrics.
- Generate predictions and analyze the results.

6. Implementation

The project was implemented in Python using the Scikit-learn library. Data preprocessing and model training were carried out in a structured pipeline to ensure reproducibility.

7. Results

The classification model successfully predicted the target classes. Performance was evaluated using metrics such as Accuracy, Precision, Recall, and F1-Score.

8. Visualizations

Recommended inclusions for your documentation:

- Dataset preview
- Training process logs
- Model performance output
- Confusion Matrix visualization
- Classification Report summary

9. Challenges

- Data preprocessing workflows
- Effective feature selection
- Model hyperparameter tuning
- Strategies for handling missing data

10. Conclusion

This project demonstrates the end-to-end implementation of a Machine Learning Classification model. The workflow includes preprocessing, training, evaluation, and prediction, providing practical experience with supervised machine learning techniques.

GitHub Repository: <https://github.com/DhanavathNavadeep/>

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